

BACHELOR'S THESIS COMPUTING SCIENCE

Detecting Hallucinations in LLM Summaries of Local Music Venues: A Hybrid Cross-Lingual NLI and Knowledge Graph Architecture

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Abstract

Large language models (LLMs) are increasingly deployed in automated content pipelines, yet remain prone to hallucination, particularly when processing niche, localized information about entities underrepresented in their training data. This thesis addresses hallucination detection in one-sentence LLM-generated summaries of local music venue events, a domain where popularity bias renders both generative models and "LLM-as-a-judge" evaluation approaches unreliable.

We propose and evaluate a hybrid two-phase evaluation architecture that decouples reading comprehension from world knowledge verification. In the first phase, a cross-lingual Natural Language Inference model (XLM-RoBERTa) evaluates semantic entailment between English atomic claims, extracted via a deterministic spaCy dependency parser, and their Dutch source texts, detecting intrinsic hallucinations without an intermediate translation step. In the second phase, an external knowledge graph (MusicBrainz) provides deterministic, non-parametric overrides for extrinsic hallucinations that the NLI model alone cannot resolve.

Evaluated on 47 newsletter summaries from the Nijmegen venue Doornroosje, the baseline NLI pipeline achieved a sentence-level F1-score of 0.8615 and a perfect error detection rate of 1.0000, though at the cost of a high false positive rate of 0.4737. Knowledge graph integration successfully reclassified 5 atomic claims from extrinsic to intrinsic contradictions with 60% accuracy, demonstrating particular strength in verifying geographic origin claims. However, due to strict min-pooling aggregation, these atomic corrections did not propagate to sentence-level improvements.

These findings demonstrate that a deterministic hybrid NLI and knowledge graph architecture is a methodologically sound alternative to circular LLM-based evaluation for niche, localized entity information, while identifying knowledge graph scope and aggregation strategy as the primary bottlenecks for future improvement.

The full Github Repository can be found at <https://github.com/rkoelewijn/bachelor-thesis>

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Chapter 1

Introduction

In recent years, large language models (LLMs) have emerged and evolved rapidly [20]. While initially restricted to research environments, these tools have now reached massive public scale. For instance, OpenAI recently confirmed it has surpassed 50 million paying subscribers across its commercial tiers, with up to 900 million weekly active users processing over 2 billion prompts per day [13]. This scale demonstrates that interacting with LLMs has become a routine global practice. However, while these models assist users in learning, working, and automating tasks, their widespread deployment does not come without downsides.

One primary risk is *hallucination*, a phenomenon further discussed in Section 2, where an LLM generates output that is unfaithful to its original source or the user’s instructions. While minor discrepancies often go unnoticed, severe hallucinations can have profound consequences. For example, Zhao et al. [21] analyzed 2.5 million published papers and found a sharp rise in non-existent references following widespread LLM adoption, conservatively estimating over 146,000 hallucinated citations in 2025 alone. Similar impacts have been documented across other critical sectors. In healthcare, hallucinations pose severe ethical and safety risks [4], while in the commercial sector, the use of ungrounded conversational agents has led to concrete operational liabilities for enterprises [8].

As mentioned above, the occurrences of hallucinations are well-documented in high-stakes fields like medicine and law. In this thesis, however, we will consider a use-case which is less universally known, a case on a local music venue in Nijmegen. Since local venues often give the stage to smaller, lesser-known artists, the likelihood of hallucinations increases. While large language models demonstrate impressive factual recall for globally prominent topics, their reliability degrades significantly when processing niche or localized information. This limitation, often referred to as popularity bias, is formally quantified by Kandpal et al. [9], who demonstrate that an LLM’s ability to accurately state a fact is directly correlated to the frequency of that entity’s appearance in its pre-training corpus.

To address these widespread hallucinations, a common modern approach is the “LLM-as-a-judge” solution, wherein a highly capable model (such as GPT-4) is prompted to verify the factual accuracy of generated text. However, recent benchmarking reveals critical structural flaws in this methodology. Chen et al. [2] introduced the FELM benchmark, demonstrating that LLM evaluators struggle significantly to detect factual errors, often failing to identify hallucinations or confidently hallucinating false corrections themselves. This failure occurs because an LLM acting as a judge relies on the same flawed, popularity-biased parametric memory as the model generating the text, where a parametric memory refers to a model’s internal knowledge determined by the model’s internal knowledge. If an LLM lacks the training data to accurately describe

a local Dutch punk band, it also lacks the data required to fact-check a summary about that band. Consequently, relying on an LLM to evaluate long-tail cultural metadata creates a circular validation flaw.

To solve this, a new approach to automated evaluation is required. Traditional hallucination benchmarks often evaluate models against static, global datasets and primarily focus purely on English tasks. These traditional measures were not set up with testing small scale musical knowledge in mind, where the source text differs from the summarisation.

Therefore, we propose a two-phase hybrid evaluation architecture designed specifically for this domain.

The intuition behind this solution is to decouple the evaluation of reading comprehension from the evaluation of world knowledge. First, to assess whether the LLM faithfully translated and summarized the provided text, we utilize a cross-lingual Natural Language Inference (NLI) model (XLM-RoBERTa). This isolates intrinsic hallucinations. Second, to overcome the LLM’s popularity bias regarding long-tail entities, we integrate a non-parametric, structured external database (MusicBrainz). By deterministically cross-referencing the LLM’s claims against this real-world knowledge graph, we can catch extrinsic hallucinations that the NLI model alone would miss. Using manually labelled claims, we compare the accuracy of our built pipeline and are able to identify wide range of hallucinations types.

Our primary research question entails:

- RQ 1** To what extent can a hybrid evaluation framework, integrating deterministic claim decomposition, cross-lingual Natural Language Inference (NLI), and a structured external knowledge graph, effectively detect and categorize hallucinations in one-sentence LLM-generated local music venue summaries?
- RQ 1.1** How accurately can a cross-lingual NLI baseline detect and classify intrinsic hallucinations by evaluating semantic entailment between English atomic claims and their Dutch source texts, and to what extent does it label extrinsic hallucinations with a neutral or supported verdict?
- RQ 1.2** How does the calibration of the baseline confidence threshold ($\mathcal{T}$) impact the models error detection and accuracy?
- RQ 1.3** How effectively can the integration of the MusicBrainz external knowledge graph increase the accuracy of atomic and sentence level labels of the baseline by validating extrinsic hallucinations?
- RQ 1.4** How does aggregating these atomic claim-level evaluations using min-pooling impact the error detection and false positive rates at the sentence level?

The remainder of this thesis is structured as follows. Chapter 2 outlines the theoretical background regarding hallucinations in LLMs, cross-lingual NLI, Knowledge Graphs and Binary Classification Metrics. Next, Chapter 3 discusses the used dataset, how it was generated and what existing problems occur. Following this, Chapter 4 discusses the full evaluation pipeline architecture, including generating the corpus, extracting source data and the external knowledge graph, parsing Dutch source text, decomposing the summaries into atomic claims, validating and evaluating these atomic claims and lastly the experiment set up. Chapter 5 shows the results of our experiments, which will then be discussed in Chapter 6 and future work will be suggested in addition to our limitations. Finally, Chapter 7 concludes the research by answering the research questions according to our results and discussion.

Chapter 2

Preliminaries

In this chapter, we introduce the concepts behind this thesis. First, we discuss the definitions of hallucinations in large language models. Secondly, we explain the subject of cross-lingual natural language interference and its relevance to this thesis. Lastly, we discuss knowledge graphs and its usage of non-parametric memory.

2.1 Hallucination in Large Language Models

In this thesis, we will be discussing the occurrences of hallucinations in LLMs. LLMs are massive artificial neural networks pre-trained on vast amounts of text data, designed to understand and generate human language. They show strong understanding in natural language and solving complex tasks [20]. While these outputs often contain useful and correct information, they are prone to *hallucinating*. Hallucination of a LLM consists of the model generating content that is not faithful to its source. Multiple types of hallucination exist, and following the taxonomy proposed by Huang et al. [6], we can categorise these into two types:

- **Factuality Hallucinations**

Discrepancies between generated content and verifiable real-world facts. This includes *Fabrications* (made-up claims) and *Contradictions* (statements that directly oppose reality).

- **Faithfulness Hallucinations**

Divergence from the provided user input or context. This includes *Instruction* (following the wrong instructions from the prompt), *Context* (Not understanding the context) and *Logical* (Incorrect logic, e.g. math errors). In information extraction (IE) tasks, these often manifest as “Missing Spans” or “Incorrect Types” [5].

Classifying the hallucinations allows us to have insight on what went wrong and where to improve, giving us the ability to do a system-level diagnosis of the pipeline [16]. For instance, a *faithfulness* error indicates a failure to maintain “context-boundary” stability [16], where a *factuality* error might need validation by means of an external source. Because LLMs are often unable to faithfully detect their own factual errors [2], deterministic verification is required to ensure airtight verification.

To understand why these factual hallucinations occur so frequently in localized domains like local music venues, we must look at how LLMs store information. LLMs encode knowledge within their network weights during pre-training, forming what is known as *parametric memory* [11]. While parametric memory may be reliable for globally prominent entities (e.g., world-famous

pop stars) due to their massive presence in the training data, it is inherently weak for niche, local entities. This limitation is known as *popularity bias*. Kandpal et al. [9] mathematically demonstrate that an LLM’s ability to accurately recall a fact drops immensely for “long-tail” entities that appear infrequently in its training corpus. Consequently, when an LLM is tasked with summarizing text about a local grassroots band, it encounters a void in its parametric memory and is prone to hallucinating plausible, fabrications in-line with what it knows about other grassroots bands.

2.2 Cross-Lingual Natural Language Inference

Natural Language Inference (NLI), also known as Recognizing Textual Entailment (RTE), is a core task in Natural Language Processing (NLP) [3]. NLP is a multidisciplinary field at the intersection between Computing Science, Linguistics, and Artificial Intelligence, where scientists focus on helping computers understand natural language. Common applications consist of machine translation, sentiment analysis, and automated summarisation. The latter is highly relevant for this thesis, since we deal with Information Extraction (IE), specifically extracting text and entities from websites [5], and Semantic Validation, which involves logically validating the semantic meaning of different sentences to detect errors [6, 2].

The task of NLI in NLP is determining the logical relationship between two pieces of text: a *premise* and a *hypothesis* [3]. Based on the truth of the premise, the model assigns one of three categorical labels as demonstrated in Table 2.1:

Table 2.1: Examples of Natural Language Inference (NLI) relationships.

Premise	Hypothesis	NLI Label
Frank Boeijen is from Nijmegen.	Nijmegen is the hometown of Frank Boeijen.	<i>Entailment</i>
Frank Boeijen is from Nijmegen.	Frank Boeijen is from Amsterdam.	<i>Contradiction</i>
Frank Boeijen is from Nijmegen.	Frank Boeijen is coming back to Nijmegen.	<i>Neutral</i>

- **Entailment:** The hypothesis is guaranteed to be true if the premise is true. In Table 2.1, the premise logically necessitates that Nijmegen is Boeijen’s hometown [3].
- **Contradiction:** The hypothesis is guaranteed to be false if the premise is true. Since the premise establishes Nijmegen as the origin, a claim of Amsterdam constitutes a direct contradiction [3].
- **Neutral:** The truth of the hypothesis cannot be determined from the premise alone. While Frank Boeijen may be “coming back” to Nijmegen, the original premise regarding his origin does not provide sufficient information to confirm or deny a return trip [3].

In the context of this thesis, we utilize *Cross-Lingual* NLI to bridge the language gap between Dutch source text and English generated summaries. By using a multilingual model like XLM-RoBERTa, we can evaluate a Dutch premise against an English hypothesis without an intermediate translation step, preserving the semantic nuances of the original concert metadata [3].

2.3 Knowledge Graphs and Non-Parametric Memory

To counteract the unreliability and popularity bias of an LLM’s internal parametric memory, modern NLP pipelines often integrate *non-parametric memory*-external, deterministically structured databases [11]. A primary form of non-parametric memory is a Knowledge Graph (KG).

A Knowledge Graph is an organized data structure that represents real-world entities as nodes (e.g., artists, venues, or albums) and the strict relationships between them as edges (e.g., “originates from” or “performs genre”). Unlike parametric models, which recall facts via learned statistical associations that degrade for low-frequency entities, a KG retrieves facts through direct structured queries with deterministic outputs.

In order to replicate the pipeline structure with an integrated non-parametric for our niche local music knowledge, open-source databases such as MusicBrainz exist. The MusicBrainz database can be implemented as a knowledge graph, allowing us to cross-reference the generated claims against verified information. This option gives us the possibility to move on from “Neutral” to a possible “Contradiction” when the source text does not provide enough context.

2.4 Binary Classification Metrics

In data mining and machine learning, evaluating the performance of a classification model requires mapping its predicted class labels against the actual ground-truth labels. For a binary classification problem, instances are categorized into a positive class or a negative class. The results are summarized in a confusion matrix [15], which tabulates the outcomes into four fundamental categories:

- **True Positive (TP):** The model correctly predicts the positive class.
- **True Negative (TN):** The model correctly predicts the negative class.
- **False Positive (FP):** The model incorrectly predicts the positive class (Type I error).
- **False Negative (FN):** The model incorrectly predicts the negative class (Type II error).

From these foundational counts, several theoretical metrics are derived to quantify different dimensions of the classifier’s predictive performance.

2.4.1 Accuracy, Precision, and Recall

Accuracy is the most widely used metric and measures the overall proportion of correct predictions made by the model. However, accuracy can be a misleading indicator when the class distribution is highly imbalanced, as a classifier can achieve a high accuracy simply by predicting the majority class [15].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.1)$$

Precision evaluates the exactness of the classifier. It represents the fraction of records that actually belong to the positive class out of all the records the model classified as positive.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.2)$$

Recall (also referred to as Sensitivity or the True Positive Rate) evaluates the completeness of the classifier. It measures the fraction of actual positive records that the model successfully detected.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.3)$$

2.4.2 F1-score and False Positive Rate

Because classification models often face a trade-off between maximizing precision and maximizing recall, the **F1-score** (or F-measure) is used to combine both into a single metric. It is calculated as the harmonic mean of precision and recall. Using the harmonic mean instead of an arithmetic mean ensures that the metric heavily penalizes a model if either precision or recall is exceptionally low [15].

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.4)$$

Finally, the **False Positive Rate (FPR)** measures the proportion of actual negative instances that were falsely classified as positive. Tracking this metric is critical in cost-sensitive learning environments where committing a Type I error carries a significant penalty [15].

$$FPR = \frac{FP}{FP + TN} \quad (2.5)$$

Chapter 3

Cloudspeakers Concert Pipeline

In this chapter, we will thoroughly discuss the architecture of the LLM summary we will be validation in this thesis, to give context on design choices made in later steps and chapters. The LLM we will be verifying was set up by Chris Bol [1].

3.1 Cloudspeakers Pipeline Architecture

Cloudspeakers is an Concert Event Pipeline setup by Chris Bol. It is an automated pipeline that collects concert events from Dutch music venues, extracts structured data using a Large Language Model, and distributes a weekly email newsletter to subscribers [1]. The system is focused on Utrecht-based venues such as ACU, Ekko, De Helling, db's, De Nijverheid, and Tivoli Vredenburg, but most importantly also contains the Nijmegen-based venue Doornroosje. The pipeline runs on Windmill, an open-source workflow automation platform. All logic is written in Python and SQL. No manual data entry is required.

The pipeline looks as follows:

1. Scrape event URLs from venue calendar pages
2. Store scraped pages (HTML + Markdown) in MinIO object storage
3. **LLM extracts structured event data (DeepSeek)**
4. Structured data stored in PostgreSQL
5. Email newsletter to subscribers

Feedback is collected from users via a web form and stored in PostgreSQL.

Step 3, LLM extraction and summarisation by the DeepSeek model, is what interest us the most. Even though we know the model and set up of the extraction, throughout this thesis the LLM and creation of the summary will be treated as a “black box”. The fields extracted by the LLM can be found in table 3.1.

Table 3.1: Description of extracted event metadata fields.

Field	Description
event	Name of the event
venue_name	Name of the venue (or hall)
main_act	Main performing artist or band
support_acts	Support acts (comma-separated)
date	Event date in YYYY-MM-DD format
date_sale	Ticket sale start date
url_sale	Ticket purchase URL
event_type	Category: gig, club night, open podium, talk, film, show, exhibition, other
genres	Music genres (comma-separated, lowercase)
summary	One-sentence English summary including artist origin
related_bands	Similar artists mentioned on the page

Table 3.1 shows that DeepSeek is instructed to generate a one-sentence English summary including the origin of the artist, which will be used to as the to be validated summary for this thesis. Other relevant field include the `main_act` which is used to in generating atomic claims in Section 4.4 and `url_sale` which is used find the source text of the summary in Section 4.1. We limit our scope to verifying the `summary`, but investigative work shows that there exist occurrences of the LLM hallucinating on the `related_bands`, where all bands mentioned on the page get marked as related, even though this might not be the case. Other fields, will be discarded in our own validation steps.

3.2 Extracted Newsletters

For our tests, we will be using 47 different output examples, of events taking place on June 10 2026 untill April 15 2027. Listing 3.1 shows one example.

```

1  act: Frank Boeijen
2  date: 2026-08-23
3  url: https://doornroosje.nl/event/frank-boeijen-19/
4  venue: Openluchtteater De Goffert
5  bands: Frank Boeijen (main)
6  genres: pop
7  summary: Dutch singer-songwriter Frank Boeijen returns to his beloved
           hometown of Nijmegen for an evening of poetic songs and stories.
```

Listing 3.1: Example of a newsletter snippet.

Since current scope of the thesis does not include validating the `type`, `date`, `venue`, `genres`, we do not need to store these. The final newsletter is stored as a JSON object, allowing our pipeline easy access. Listing 3.2 shows a complete generated entry.

```

1 {
2   "event": "Daoud Live",
3   "venue_name": "Doornroosje",
4   "main_act": "Daoud",
5   "genres": ["jazz", "hip-hop"],
6   "summary": "French trumpeter Daoud is bringing his unique blend of jazz
7               and hip-hop to Nijmegen for the first time."
8 }

```

Listing 3.2: Example of a simplified newsletter entry.

All used newsletters can be found in Appendix A.1.

3.3 Discussing Historical Feedback

At the bottom of each email sent by the pipeline, a link is put with a request for feedback to the reader. To see whether hallucinations already occurred during this pipeline, we requested the full feedback list from Bol.

The feedback list, contains 11 different feedback entries, of which three are suggestions of new features, two entire are about the **genre** being labelled incorrectly, two entries where the act (main vs supporting) is labelled wrong and lastly, two actual hallucinations.

The hallucination feedback reads: “I wonder whether *Luna Maki* really is from Utrecht.” A short online search shows that Luna Maki is actually from Amersfoort¹.

The second person notes: “*Naaz* genre not great. Text says ‘unique sound’ but the LLM turns it into regular folk music.” This claim was not easily confirmed, but still shows us that the LLM has problems with defining the origin and might be interpreting genres wrong.

¹<https://www.nieuwsplein33.nl/nieuws/3867624/luna-maki-op-het-punt-van-doorbreken-met-afrobeat-uit-amersfoort>

Chapter 4

Summary Evaluation Pipeline Architecture

This chapter explains the pipeline structure set up to validate the LLM summaries generated for the Cloudspeakers Newsletter. The pipeline consists of five modules, building the dataset, decomposing the summary into atomic claims, parsing the Dutch source text, evaluating validity of the claims against the source text and external database knowledge from MusicBrainz and lastly evaluating the outcome.

In order to correctly verify the summary given by the LLM model, we have to create a pipeline that sources the needed context. As seen in Chapter 3, the newsletter contains a one-sentence summary. Before we can validate whether the claim is supported, we need to extract all context. This is done by loading the URL using a headless browser, where we scrape the Dutch text. This then gets parsed and cleaned, and sent to be validated by a NLI model. Meanwhile, the summary gets separated into atomic claims, to be validated. The NLI model consists of a Transformer-based language model which can perform cross-lingual comparisons, which compares the source, our premise, to the hypothesis, atomic claims. This pipeline follows the *segmentation, matrix generation, aggregation* structure suggested by Laban et al. [14]. However, in our experiment, we supply our baseline evaluation with the external knowledge graph, to aid the program in finding and correcting hallucinations. We also do not use any of the mean aggregation methods, but instead opt to use *min aggregation*. Using entailment scores of our claims, we are able to classify two types of hallucinations. A diagram showing the full framework can be found in Figure 4.1.

Each paragraph will explain in depth the architectural choices that were made and how each module of the pipeline works.

4.1 Corpus Construction

Before any validation or evaluation is possible, we need to load the source data given in the newsletters' URL field. This allows us to create a so-called *corpus* which contains all available data we have on the artist we are evaluating. The corpus setup is implemented by loading all the data up front, so we save time loading all data again when our pipeline crashes later on, or if we wish to run the experiment again. This process collects the data in three simple steps: loading the local venue data, fetching the MusicBrainz facts, and saving everything to a JSON file.

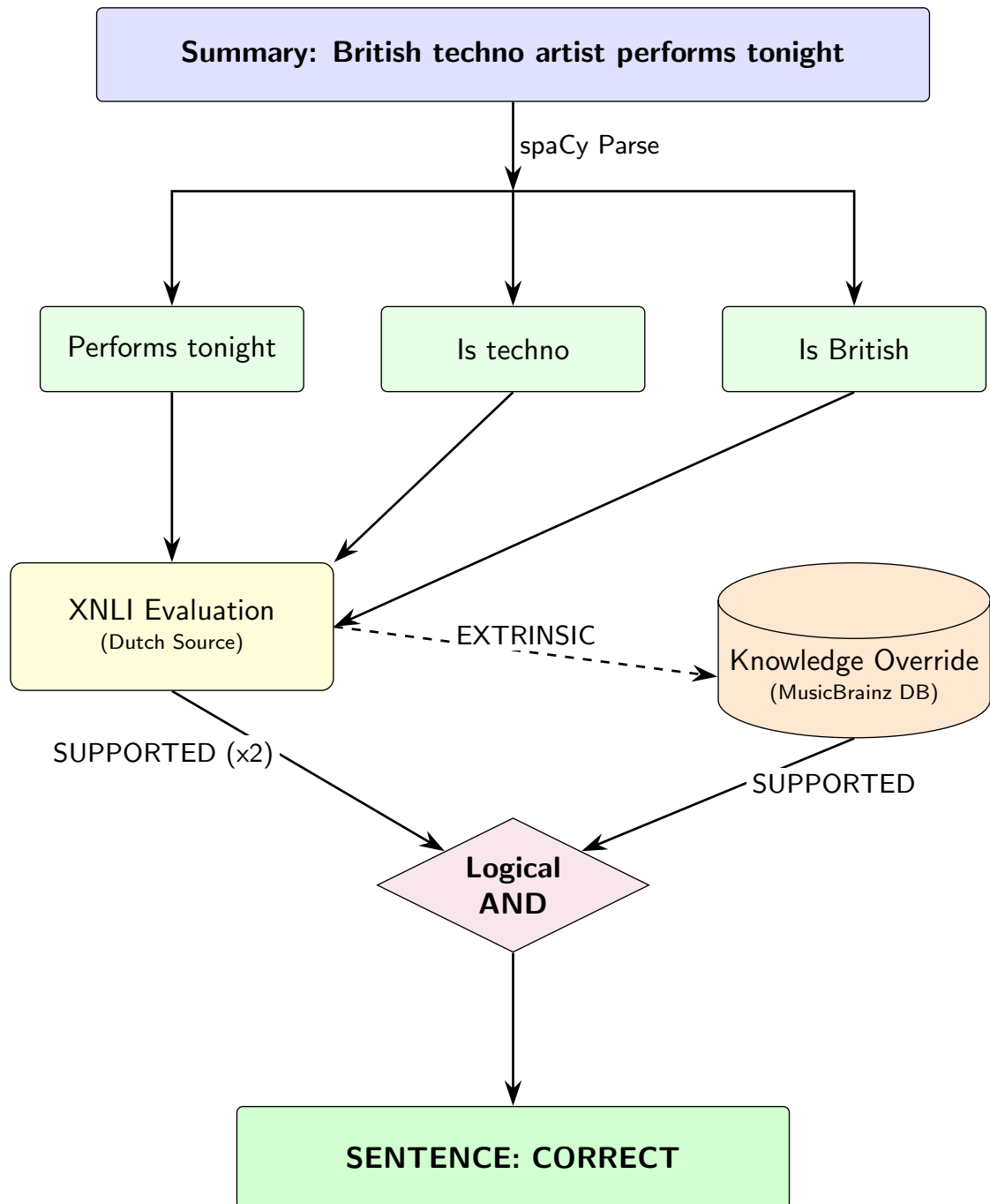


Figure 4.1: System architecture flowchart demonstrating the corpus ingestion, atomic claim decomposition branches, and cross-lingual source verification steps. Blue nodes represent the cross-lingual NLI evaluation path; the dashed red branch indicates the extrinsic knowledge override via MusicBrainz; green nodes represent final verdicts.

4.1.1 Extracting Venue Data

As mentioned in Chapter 3, our primary data source is a generated newsletter, where we will be focussing on the venue Doornroosje, located in Nijmegen. This is set up using a script, where the primary event details and the LLM-generated summaries are manually extracted from these newsletters and provided to the script.

The events page gets opened using a chromium headless browser, set up using the **Playwright** package. This is done since we wanted to ensure that our web scraping was done in a dynamic and deterministic fashion and we could replace the venue if we wanted to. This meant the option of using an AI Webcrawler was out of the equation, due to it not being deterministic and it being too costly to set up. To prevent memory leaks and segmentation faults during the batch processing, we allocated a single browser context and passed every URL field from the input newsletters iteratively.

Once we collected the HTML using the headless browser, we used **bs4** from the **BeautifulSoup** package to find the **main**, **body** or **article** which contained the text originally scraped by the LLM. Other HTML tags were discarded. This source text is then passed on to our Dutch Parser module, which parses, cleans and prepares the Dutch text to be used as the premise of the NLI check.

4.2 External Knowledge Graph Implementation

The external knowledge graph we chose for this experiment is the MusicBrainz Database [12]. MusicBrainz is a community-maintained, open source encyclopedia of music information. It stores all of the music information that is collected by users, from artists and their releases to works and their composers. MusicBrainz gives us the availability to use this service by calling their API, allowing us to automatically request information.

4.2.1 MusicBrainz Data Extraction

Once the primary event data is manually set up, the script automatically fills the dataset with verified real-world facts by querying the MusicBrainz API. To respect the strict rate-limiting policies enforced by the MusicBrainz infrastructure, our script pauses for a second, before querying the next artist.

For every artist identified in the venue data, the script searches the API to retrieve three specific data points: **name**, **type**, **country**, **area**, **place** and **genre**. A description and example of each field can be found in Table 4.1.

Field	Description	Example
name	The official name of the artist, band, or act.	Frank Boeijen
type	The classification of the entity (e.g., Person, Group).	Person
country	The primary country of origin for the artist.	NL
area	The broader geographical region or country.	Netherlands
place	The specific founding city or hometown (begin-area).	Nijmegen
genre	A comma-separated list of the top two user-generated genre tags.	singer-songwriter

Table 4.1: Description and examples of factual metadata fields extracted from the MusicBrainz API.

Given that mid-tier venues frequently program niche, emerging, or local artists, it is expected that some acts will be entirely absent from the MusicBrainz database. To ensure the robustness of the dataset generation, the script implements explicit error handling. If an artist cannot be resolved, the pipeline catches the exception and assigns empty values to the factual fields rather than terminating the program, allowing the script to safely proceed to the next entity.

4.3 Parsing Dutch Source Text

The main task of this module is to clean the extracted text, since HTML source text delivered by the corpus module contains boilerplate text that can clutter up our analysis. One factor that we noticed, was a constant section of a cookies pop-up text. During test runs of the web scraping, our corpus was filled with the paragraph that can be found in Listing 4.1.

```
1 We gebruiken cookies om je ervaring op onze website te verbeteren. Lees
  hier ons privacybeleid.
2 Noodzakelijke cookies zijn cruciaal voor de basisfuncties van de website
  en zonder deze werkt de website niet op de beoogde manier.
3 Deze cookies slaan geen persoonlijk identificeerbare gegevens op.
4 Functionele cookies helpen bepaalde functionaliteiten uit te voeren,
  zoals het delen van de inhoud van de website op sociale mediaplatforms
  , het verzamelen van feedback en andere functies van derden.
5 Analytische cookies worden gebruikt om te begrijpen hoe bezoekers omgaan
  met de website. Deze cookies helpen informatie te verstrekken over de
  statistieken van het aantal bezoekers, het bouncepercentage, de
  verkeersbron, enz.
6 Prestatiecookies worden gebruikt om de belangrijkste prestatie-indexen
  van de website te begrijpen en te analyseren, wat helpt bij het
  leveren van een betere gebruikerservaring voor de bezoekers.
7 Advertentiecookies worden gebruikt om bezoekers gepersonaliseerde
  advertenties te bezorgen op basis van de eerder bezochte pagina's en
  om de effectiviteit van de advertentiecampagne te analyseren.
8 Andere niet-gecategoriseerde cookies zijn cookies die worden geanalyseerd
  en die nog niet in een categorie zijn ingedeeld.
```

Listing 4.1: An example of boilerplate cookie consent text frequently extracted as noise during the initial web scraping phase.

Instead of filtering this text statically, which in theory would work since we are only scraping from a single source, we opted for a more dynamic approach to ensure that when this pipeline is adapted to another webpage less work is needed. This dynamic approach is heuristic-based, which works since basing our rules on shallow text features for boilerplate removal achieves high accuracy without the need for complex structural features or site-specific engineering [10]. Our code utilizes regular expressions (regex) to search and remove Listing 4.1. First, we define a list of patterns, which are the commonly found phrases in cookie banners, such as “We gebruiken cookies” and “Functionele cookies”. Then, we match the rest of the sentence and loop through the document and replace any matching string with an empty string, deleting the snippet. Lastly, we clean up large chunks of white space.

After having removed the boilerplate text from the source, we need to append context about the artist to each sentence. As the validation is done over every single source line, context can be lost due to the artist name not being in the sentence. To solve this issue, we prepend the following string: “Over artist_name:”, where the name of the artist is inserted. This addition

ensures that the sentences keep their main focus, which is a necessity for the validation process in Section 4.5. The clean, contextualised Dutch source sentences are then passed on to the validator. All these individual sentences form the source sentence set $\mathcal{S} = \{s_1, s_2, \dots, s_n\}$.

4.4 Decomposing Summaries Into Atomic Claims

The main task of the decomposer module is to separate the claims made by the LLM in the generated summary. A summary can contain from one until multiple claims, so it is important that all of them get separated into *atomic claims* and get evaluated. We define a claim as *atomic* if it contains exactly one predicate-argument relationship, which allows the subsequent NLI model to render a definitive hallucination label without semantic noise. This falls back on the first step of Laban et al.’s structure, which exists of *segmentation*, where we decompose a summary $\mathcal{C}$ into atomic claims $\mathcal{C} = \{c_1, c_2, \dots, c_m\}$.

4.4.1 Decomposing Intuition

While state-of-the-art frameworks like Google DeepMind’s SAFE [17] rely on LLM prompting to decompose long-form summaries into atomic claims, introducing yet another machine learning model and creating high operational and computational costs, we made the choice to do this in a deterministic fashion. Our methodology adopts the core multi-stage atomic decomposition paradigm pioneered by SAFE, but replaces the LLM parser with a deterministic, rule-based linguistic dependency parse utilizing `spaCy`. Splitting claims into atomic claims, has been proven to improve performance in NLI for multiple models [7, 14]. Our decomposer has been developed in parallel to the framework given by Huang et al. [7].

By utilizing a deterministic linguistic dependency parser rather than a secondary LLM for claim decomposition, we avoid the ‘recursive hallucination’ risk [19]. When an LLM is used to decompose a summary, it may introduce its own hallucinations during the parsing phase. Our rule-based approach ensures that the decomposition remains faithful to the text’s syntactic structure, effectively providing a stable, verifiable input for the subsequent NLI evaluation.

4.4.2 Decomposer Implementation

The core of our decomposer consists of treating a sentence as a hierarchical Dependency Tree, where words are nodes connected by functional relationships. These relationships mimic the function of words in English sentences, such as a *preposition*, *subject*, *object*. The enumeration below shows the implementation of the algorithm.

1. Convert the raw text into a machine-readable dependency tree. This allows the system to identify the grammatical roles of words, such as which terms function as the subject and which function as the object of an action.
2. Iterate through every token in the document to locate “action anchors.” The system prioritizes VERB and AUX (auxiliary verbs such as ‘is’ or ‘has’), as these words define the core events or states necessary for a factual claim.
3. Resolving Components:
 - Subject: Identify the nominal subject (‘nsubj’). If the subject is a pronoun (e.g., “He”), replace it with the primary entity name to ensure the claim remains contextually independent and verifiable.

- Verb Phrase: Collect all auxiliary verbs to preserve the original tense (e.g., “is performing”).
 - Object: Locate the grammatical receiver of the action (direct object, attribute, or complement).
 - Modifiers: Identify surrounding phrases that provide extra context, such as time, location, or manner.
4. Combine the subject, verb phrase, and object to construct a coherent, standalone sentence that represents the core fact.
 5. If the sentence includes modifiers, “multiply” the base statement by each modifier to generate individual, independent claims. This step converts one complex sentence into multiple simple, verifiable statements.
 6. Perform final data hygiene. Remove exact duplicate claims and discard any fragments that are under 5 words in length. This ensures the final output contains only high-quality, meaningful information.

The atomic claims set has now been created and can be validated by the next module.

4.5 Validating Atomic Claims Based on Source Text

Once the atomic claims have been generated and the source text has been cleaned and contextualised, we can start the validation process. The validation module of the pipeline compares the hypothesis, atomic claims, against the premises, contextualised source text, and returns three different scores, **entailment**, **contradiction** and **neutral**. Based on these scores, we can determine the validity of the claim. Since the source text differs from the atomic claims, we will be using a cross-language NLI (XNLI) model, to ensure no context is lost in a (separate) translation layer.

4.5.1 Validation Design Choices

As mentioned in Section 1, the option of using a generative model for the entailment check was discarded to avoid the inherent bias of self-validation where a LLM acts as an uncritical judge of its own generated output. Since the research objective is to establish the viability of a fully deterministic pipeline, we selected Facebook’s XLM-RoBERTa [3]. XLM-RoBERTa, according to Conneau et al., serves as an extensively documented ‘gold standard’ baseline in cross-lingual NLI literature.

XLM-RoBERTa is a Transformer-based language model trained on 2.5 Terabytes of filtered CommonCrawl data across 100 different languages. XLM-RoBERTa learns a shared mathematical representation of human language, which results in it being able to convert multiple languages into a mapped mathematical vector space, meaning different languages can be compared. Conneau et al. show that XLM-RoBERTa is very competitive with strong monolingual models on XNLI benchmarks, demonstrating that it is possible to have a single large model for all languages. [3]. For the Cloudspeakers Experiment (Section 3), this saves us steps from translating between Dutch and English which could lead to information and/or performance loss, or limiting the experiments to pages that *do* have an English variant. As mentioned in 2.2, NLI looks at the entailment, contradiction and neutral scores. XLM-RoBERTa outputs these when given a premise and hypothesis and using these we can determine likelihood of a hallucination by the

LLM. Table 4.2 shows a validation example on C.1 and C.2, only differing by the hometown of the artist. The **neutral** score is not shown, to simplify the example.

Table 4.2: An example of the validation output generated by XLM-RoBERTa, detailing the confidence scores and the extracted evidence used for each claim.

C.1: *Dutch singer-songwriter Frank Boeijen returns to his beloved hometown of Nijmegen*

Verdict	Ent. Score	Con. Score	Evidence Used
ENTAILMENT	99.95	0.11	Frank Boeijen keert samen met zijn band terug naar Openluchttheater de Goffert, in zijn geliefde thuisstad Nijmegen.

C.2: *Dutch singer-songwriter Frank Boeijen returns to his beloved hometown of **Amsterdam***

Verdict	Ent. Score	Con. Score	Evidence Used
CONTRADICTION	0.12	99.30	Frank Boeijen keert samen met zijn band terug naar Openluchttheater de Goffert, in zijn geliefde thuisstad Nijmegen.

4.5.2 XLM-RoBERTa Implementation

The **validator** is implemented in Python utilizing PyTorch and the HuggingFace **transformers** library, using the specific `joeddav/xlm-roberta-large-xnli` model. As mentioned in Section 4.4, the pipeline delivers multiple different atomic claims ($c_1^A, c_2^A, \dots, c_n^A$) for every summary $\mathcal{C}^A$ and the source sentences $S^A = \{s_1^A, s_2^A, \dots, s_m^A\}$ for every artist A . Since XLM-RoBERTa is set up to only check one premise against a single hypothesis, we will run the model for every claim and every sentence in the source material and find which sentence results in the highest entailment and which in the highest contradiction. We chose to run the experiments in this way, to adhere to the segmentation framework [14] and to remain below XLM-RoBERTa’s maximum input token capacity. To be certain we won’t go above XLM-RoBERTa’s 512-token input limit, we tokenize the pairs (c_i^A, s_j^A) with truncation enabled. This is just a precaution, because during our experiments XLM-RoBERTa never complained about the token limit being reached.

The raw numbers (logits) output by the model are passed through a Softmax function to calculate a probability distribution across the earlier mentioned classes: **contradiction**, **neutral** and **entailment**. This ensures they all add up to a value of 1.

The script holds two trackers for each claim in $\text{mathcal{C}}^A$:

1. $\text{ent_score}_{\max}(c_j^A)$: The maximum entailment score of Claim j for all sentences S^A .
2. $\text{con_score}_{\max}(c_j^A)$: The maximum contradiction score of Claim j for all sentences in S^A .

These scores and sentences are then passed to the evaluation module; together, the two trackers represent the peak values of the $M \times N$ pair matrix described by Laban et al [14] with every possible pair (c_i, s_j) . However, we are not interested in the entire matrix, but just the peaks, so

we do not store the entire matrix but just the max scores. This is due to the labelling logic will apply in Section 4.6.1.

4.6 Evaluating Validation Scores

The evaluation of the newsletter summaries, will be done in a two step process. After XLM-RoBERTa has finished generating validation scores for each claim of a summary sentence, we will first label the claims as **SUPPORTED**, **INTRINSIC** or **EXTRINSIC** [6] based on the Dutch source text. Flagging the claims will be done based on the trackers set up in Section 4.5. Next, after the hallucinations have been marked, we aggregate these results in based on the SAFE framework [18], by using a strict atomic verification, which entails that every claim of the sentence has to be marked **SUPPORTED** for a sentence to be given the **CORRECT** verdict. Otherwise, it receives **INCORRECT**.

We choose this binary evaluation approach, since we use a SummaC-based [14] baseline approach, but to also not lose the information gathered by classifying the hallucinations.

4.6.1 Ternary Atomic Claim Evaluation

To correctly map every atomic claim onto one of the three claim labels (**SUPPORTED**, **INTRINSIC** or **EXTRINSIC**), we make use of the trackers set up in Section 4.5. Equation 4.1 shows when a label is applied.

$$\text{Verdict}(c_j) = \begin{cases} \text{SUPPORTED} & \text{if } \text{ent_score}_{\max}(c_j) \geq \mathcal{T} \quad \text{and} \quad \text{ent_score}_{\max}(c_j) > \text{con_score}_{\max}(c_j) \\ \text{INTRINSIC} & \text{if } \text{con_score}_{\max}(c_j) \geq \mathcal{T} \quad \text{and} \quad \text{con_score}_{\max}(c_j) > \text{ent_score}_{\max}(c_j) \\ \text{EXTRINSIC} & \text{otherwise} \end{cases} \quad (4.1)$$

For all c_j in $\mathcal{C}$, $\text{ent_score}_{\max}(c_j)$ is that claims highest entailment score for sentence s_j , $\text{con_score}_{\max}(c_j)$ is that claims highest contradiction score for sentence s_j and $\mathcal{T}$ referring to the confidence threshold.

4.6.2 Binary Sentence Evaluation

After having given a verdict to the claims of a sentence, we start the aggregation process of these claims. The form of aggregation we have chosen is from the SAFE framework [17], where every claim must hold before a parent sentence can hold. While the SummaC [14] architecture utilizes mean-pooling to aggregate summaries, this approach risks missing hallucinations due to a sentence having a large number of claims. We opt to implement *min-pooling*:

- **CORRECT:** A sentence is labeled as factually consistent if and only if all of its atomic claims are classified as **SUPPORTED**.
- **INCORRECT:** A sentence is labeled as inconsistent if at least one of its atomic claims is classified as an **INTRINSIC** or **EXTRINSIC** hallucination.

This strict binary output serves as the final evaluation metric for our pipeline, allowing us to calculate the error detection rate when comparing the baseline model against the experiment model, which uses the external knowledge graph.

4.7 Experiment setup (Knowledge Graph Integration)

To assess whether a hybrid evaluation framework, implementing XLM-RoBERTa with the MusicBrainz integrated knowledge graph, can effectively detect and classify intrinsic and extrinsic hallucinations, we have set up our discussed framework. At the point of evaluation, we will see whether inserting the external data can improve our sentence verdict. By helping the validation module by inserting more data we confirmed holds, we hope to improve its overall score.

4.7.1 Baseline Evaluation Protocol

To compare the results, we will first gather results without the external database, to create a baseline. This baseline can be loosely compared to other NLI Summary Frameworks [14], as well as modern LLM-driven factuality evaluators [2]. For this baseline framework, we have to also calculate the optimal value for $\mathcal{T}$, mentioned in Equation 4.1. This is done after having generated the validation scores. The scores are saved, and we run a check for $0 \leq \mathcal{T} \leq 100$ in increments of 5, creating 21 unique thresholds. We choose this size of increments, since smaller increments may lead to overfitting of the model. Once the optimal threshold for our baseline setup has been calculated, we apply this threshold to the experiment setup to guarantee that possible improvements can be attributed to our addition of an external knowledge graph.

4.7.2 External Knowledge Graph Override

As we wish to insert to external knowledge graph into our structure to assess whether our hybrid framework works, we implemented an override system. Since XLM-RoBERTa is built on comparing natural language, we cannot compare the values from the MusicBrainz API directly to our hypothesis. For this, we will need to generate synthetic source text. For every field that contains data in the MusicBrainz database, we fit it into a custom template sentence. These sentences can be found in Table 4.3.

MusicBrainz Field	Synthesized Template Sentence
area	<i>[Artist] originates from [Area].</i>
country	<i>[Artist] originates from [Country].</i>
genre	<i>[Artist] plays [Genre] music.</i>

Table 4.3: Template sentences used to synthesize structured MusicBrainz data into natural language premises for cross-lingual NLI validation.

Next, we do another validation round using XLM-RoBERTa on the claims which were marked INTRINSIC or EXTRINSIC and our synthesized sentences. Using the same threshold $\mathcal{T}$, we relabel these claims and override the existing value if they differ.

4.7.3 Manual Ground Truth Labelling

To check the accuracy of our labels for claims and sentences, we have manually marked every sentence and claim, so that the results can be compared. Creating ground-truth labels for the claims, which leads into the ground-truth for the sentences, was done based on the following considerations:

- **Completeness:** Every topic mentioned in the summary had to be able to be traced back to (a part of) a sentence.
- **Factuality:** All claims had to be truthful according to the source text, meaning no contradictions were found in the source.

Notably, we based the information on the source, and not global knowledge. This entails that if we knew that e.g. “Adult DVD”, wasn’t from The Netherlands, because we might have been to one of their concerts, and the summary claimed that they were, we mark the claim as **EXTRINSIC** since the knowledge of where “Adult DVD” originates from is not present in the source material. If we were to update the source with knowledge of the “Adult DVD’s” origin, or extend it using another database, the hallucination will be labelled as **INTRINSIC**. These ground truth labels can be found in Appendix A.2.

4.7.4 Evaluation Metrics

To evaluate the performance of both the baseline and experimental pipelines, we use the following metrics:

- **Threshold Optimization:** To assign the confidence threshold ($\mathcal{T}$) we derive it by evaluating the continuous entailment scores of the baseline model against the ground truth and locking the value that maximizes the **F1-score**.
- **Atomic Claim Evaluation:** At the component level, we utilize a standard confusion matrix, alongside **Precision, Recall, and F1-scores**, to measure the cross-lingual NLI model’s ability to accurately categorize the ternary labels (**SUPPORTED, INTRINSIC, EXTRINSIC**). This level of measurement is critical for quantifying exactly how many claims are impacted by the external knowledge graph.
- **Binary Sentence Evaluation:** At the final aggregation level, where the min-aggregation assigns a definitive **CORRECT (1)** or **INCORRECT (0)** label, we evaluate the overall system using standard binary classification metrics: **Accuracy, Precision, Recall, and F1-score**.

To view the impact of the external knowledge graph integration, we emphasize two targeted metrics within this classification:

- **Error Detection Rate (Recall):** The proportion of ground-truth hallucinations successfully flagged by the pipeline.
- **False Positive Rate:** How often the pipeline falsely accuses a summary of containing an error.

Chapter 5

Results

5.1 Experimental Setup

The experiment executed in this chapter, consists of calculating a baseline measurement, by optimizing threshold $\mathcal{T}$ to find the maximum sentence level F1-score, inserting the external knowledge graph into our pipeline and overriding classifications in the baseline pipeline, hoping to improve the false positives count and error detection rate, resulting in an improved F1-score. The dataset we will be performing these experiments on contains 28 sentences which contain a hallucination, which need to be labelled **INCORRECT**, and 19 that are factual (**CORRECT**), the distribution can be found in Table 5.1. The distribution of claims can be found in Table 5.4.

Table 5.1: Distribution of ground truth labels in the binary sentence dataset.

Label	Count	Percentage
INCORRECT (Hallucinated)	28	59.6%
SUPPORTED (Correct)	19	40.4%
Total	47	100.0%

5.2 Baseline NLI Performance

In order to score the succes of our experiment, we will first need to set a baseline measurement. This baseline consists of calculating the confidence threshold $\mathcal{T}$ and running the pipeline without inserting external knowledge.

5.2.1 Confidence Threshold Optimization

To determine the label assigned to an atomic claim in our evaluation module, we use the confidence threshold ($\mathcal{T}$). Before evaluating the baseline pipeline’s predictive performance, the optimal confidence threshold ($\mathcal{T}$) was established to maximize the F1-score on the sentence level. We chose to maximize this metric, since the primary objective of the framework is to detect hallucinations at the sentence level and optimizing strictly for atomic performance worked counterintuitive. The results of our threshold optimization can be found in Figure 5.1.

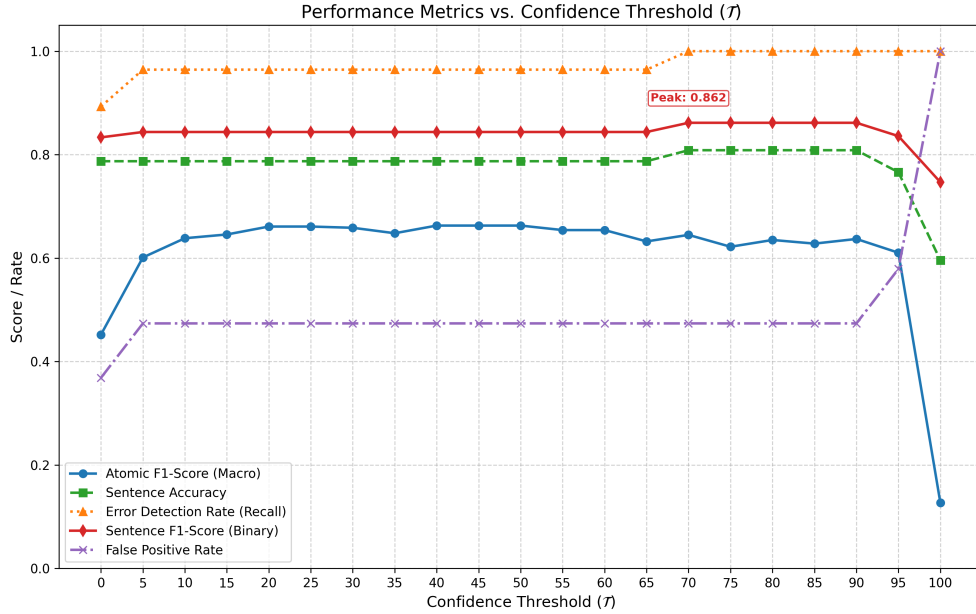


Figure 5.1: Macro F1-Score plotted against the confidence threshold ($\mathcal{T}$) during the baseline evaluation. The model’s predictive performance plateaus at exactly $\mathcal{T} = 0.65$, demonstrating the strict polarization of the XLM-ROBERTa softmax logits.

Figure 5.1 shows that the pipeline reaches a maximum F1-score of 0.862 at $\mathcal{T} = 70$. Table 5.2 shows the values of all lines at this $\mathcal{T}$. Even though the F1-score of the atomic claims is higher at $\mathcal{T} = 40$, the F1 score is higher at 70 due to the error detection rate being 1.0, entailing that every error was found correctly, and the accuracy reaching 0.8085. The high threshold does introduce a False Positive Rate of 0.4737 at the sentence level, but this rate is constant from thresholds ranging from 5 until 90.

Table 5.2: Performance Metrics at Optimal Sentence-Level Threshold ($\mathcal{T} = 70$)

Level	Metric	Score
Sentence	F1-Score	0.8615
Atomic	F1-Score (Macro)	0.6448
Sentence	Accuracy	0.8085
Sentence	Error Detection Rate (Recall)	1.0000
Sentence	False Positive Rate	0.4737

These flat scores directly follow from the distribution of confidence scores generated by the XLM-RoBERTa baseline. As seen in Figure 5.2, the model exhibits a confidence distribution where the vast majority of assigned scores cluster at the extremes ($\geq 90\%$ or $\leq 20\%$). Because the model rarely outputs intermediate confidence scores, adjusting the threshold within this massive section band doesn’t lead to a significant number of claims receiving a different classification. This behavior also explains why the variance between the lowest and highest possible F1-Scores across the non-extreme ($90 \leq \mathcal{T} \leq 20$) threshold range is a difference of only 0.04 (approximately

4%).

For our other calculations, $\mathcal{T} = 70$ was selected. Since the metrics did not improve after this $\mathcal{T}$ and by increasing the threshold we increase the possibility of our modelling being prone to be overfitted.

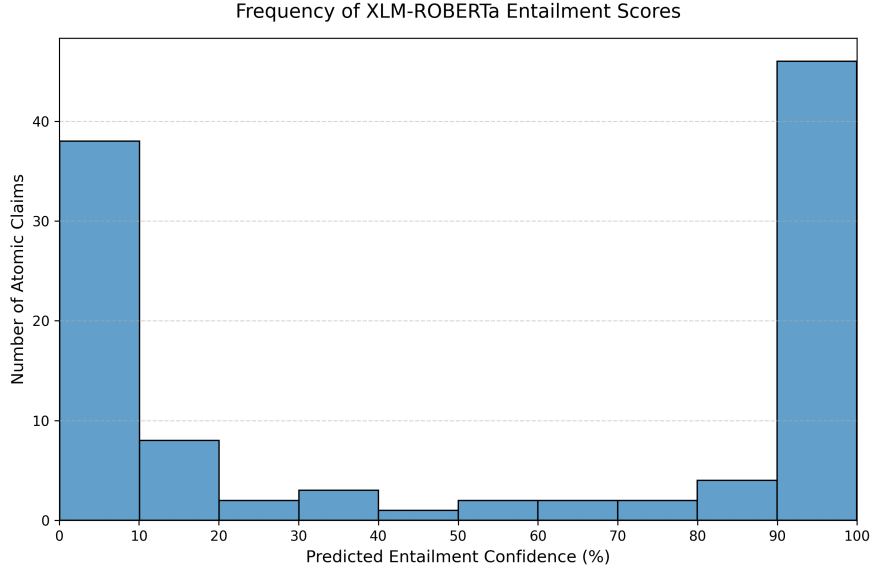


Figure 5.2: Histogram showing the distribution of model entailment confidence across all atomic claims. The bimodal pattern confirms that the majority of predictions are polarized toward total certainty or total uncertainty.

5.2.2 Atomic Claim Level Performance

The baseline performance of the XLM-RoBERTa model at the atomic claim level allows us to look at the system’s ability to assign the **SUPPORTED**, **INTRINSIC** and **EXTRINSIC** labels. As summarized in Table 5.3, the model achieved a Macro F1-Score of 0.526, with a Precision of 0.737 and a Recall of 0.601. These metrics indicate that the model scores well on the task of identifying claims that are supported by the source, but has a lower accuracy on labelling the intrinsic and extrinsic hallucinations.

Table 5.3: Performance Metrics at Optimal Sentence-Level Threshold ($\mathcal{T} = 70$)

Metric	Score
F1-Score (Macro)	0.6448
Accuracy	0.7653
Error Detection	0.8837
False Positive	0.3273

To understand the model’s classification errors, the confusion matrix in Figure 5.3 shows the distribution of our predictions across the three atomic labels. Due to the data set containing 51% of claims that are labelled **SUPPORTED** in our ground truth (Table 5.4), the matrix shows the most values for the supported labels. The model marked 12 **SUPPORTED** claims as **INTRINSIC**, this backs up the False Positive numbers we encountered.

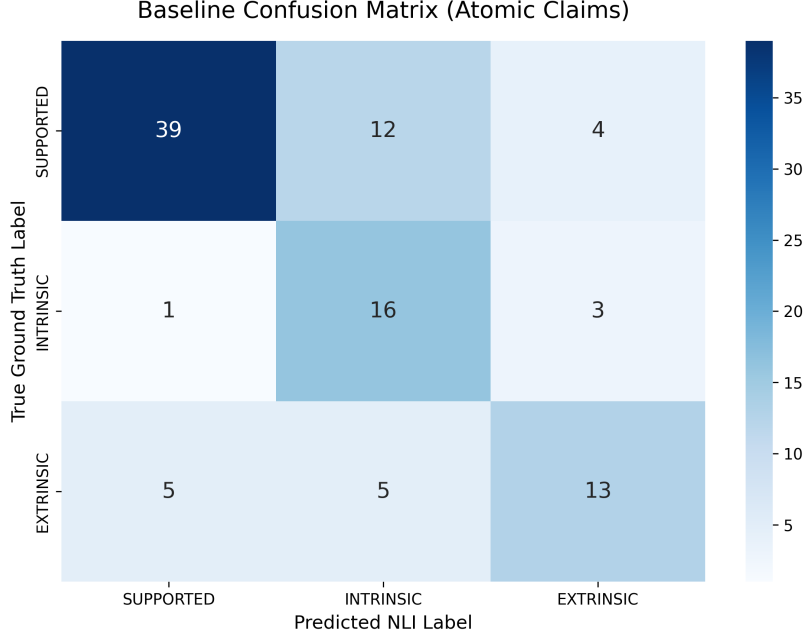


Figure 5.3: Confusion matrix of the baseline XLM-ROBERTa model at the atomic claim level. The heatmap illustrates the model’s distribution of predictions across the three label categories. The significant misclassification of ‘EXTRINSIC’ hallucinations as ‘SUPPORTED’ underscores the model’s tendency to conflate high lexical overlap with factual entailment.

Table 5.4: Distribution of Ground Truth vs. Predicted labels in the atomic claim dataset at Baseline ($\mathcal{T} = 70$).

Label	Ground Truth		Predicted (Baseline)	
	Count	Percentage	Count	Percentage
SUPPORTED	55	56.1%	42	42.9%
EXTRINSIC	24	24.5%	25	25.5%
INTRINSIC	19	19.4%	31	31.6%
Total	98	100.0%	98	100.0%

As detailed in Figure 5.3, the most frequent error was the misclassification of 12 SUPPORTED claims as INTRINSIC. Additionally, 5 EXTRINSIC claims were incorrectly labeled as SUPPORTED, and 5 EXTRINSIC claims were labeled as INTRINSIC.

5.2.3 Sentences

Using the min aggregation method from Section 4.6, we are able to use the threshold $\mathcal{T} = \iota$ to classify sentences as either CORRECT or INCORRECT.

In Table 5.5, the final baseline metrics can be found. We will use these values as a baseline to check our experiment against. Notable is that no False Negatives were found, meaning that we did manage to detect every sentence with a hallucination. However, this does come with the downside of our False Positive Rate reaching 0.4737, meaning that our baseline model flags almost half of verifiable sentences as a hallucination.

Table 5.5: Baseline performance metrics aggregated at the sentence level.

Metric	Score
F1-Score	0.8615
Accuracy	0.8085
Error Detection	1.0000
False Positive	0.4737

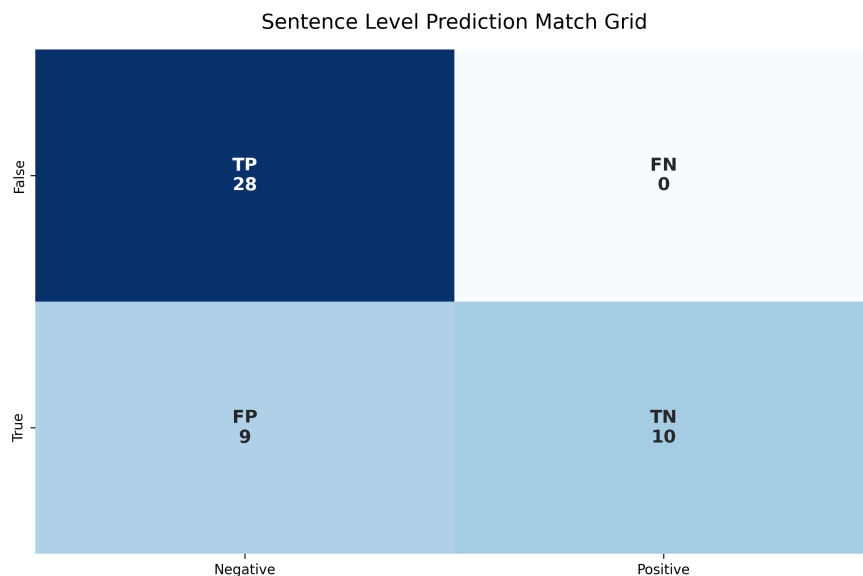


Figure 5.4: Sentence-level prediction match grid (confusion matrix) for the baseline pipeline at $T = 70.0$, showing 28 True Positives, 0 False Negatives, 9 False Positives, and 10 True Negatives.

5.3 Knowledge Graph Experiment Results

The goal of the experimental set up is to test if injecting the MusicBrainz data, an external knowledge graph, can improve the models performance, based on the baseline sentence level F1-score (0.8615) and the baseline false positives rate (0.4737). During the baseline set up, the confidence threshold of $\mathcal{T} = 70$ was established and will be used for our calculations, to ensure that both pipelines are identical apart from the addition of the knowledge graph.

The knowledge graph is able to insert data ranging over the following topics: **area**, **country** and **genre**, which can be found in Table 4.3. In total, 56 claims (see Table 5.4) were attempted to be overridden, since we only try to change verdicts that are **INTRINSIC** or **EXTRINSIC** and the result will either be **INTRINSIC** or **SUPPORTED**.

5.3.1 Experiment Atomic Claim Level Performance

At the atomic claim level, the integration of the Knowledge Graph resulted in a shift exclusively within the ternary metrics. As shown in Table 5.6, the Macro F1-Score decreased slightly to 0.6153. This minor decrease occurs because the ternary calculation treats all three labels as distinct categories; moving a claim from one error class to another alters the distribution without improving the binary correct/incorrect ratio. To understand this shift, we examine the specific label overrides.

Table 5.6: Experimental Atomic Performance Metrics at Optimal Sentence-Level Threshold ($\mathcal{T} = 70$)

Metric	Score
F1-Score (Macro)	0.6153
Accuracy	0.7653
Error Detection	0.8837
False Positive	0.3273

As we can see in Table 6.2, 5 **EXTRINSIC** claims have been overwritten to be **INTRINSIC**. Of the 5 overrides, 3 involved geographic origin claims and 2 involved genre claims. Accuracy was higher for origin claims (3/3 correct) than for genre claims (0/2 correct), as seen in Table 5.7.

Table 5.7: Summary of all Knowledge Graph interventions ($N = 5$). The table details the semantic category of the synthesized premise and whether the resulting hybrid override matched the human-annotated Ground Truth.

Artist	Attribute Category	Base Verdict	Hybrid Verdict	Ground Truth	Match?
Adult DVD	Origin	EXTRINSIC	INTRINSIC	INTRINSIC	✓ Yes
Chica Chica	Origin	EXTRINSIC	INTRINSIC	INTRINSIC	✓ Yes
HENGE	Origin	EXTRINSIC	INTRINSIC	INTRINSIC	✓ Yes
daoud	Genre	EXTRINSIC	INTRINSIC	EXTRINSIC	× No
Daryll-Ann	Genre	EXTRINSIC	INTRINSIC	EXTRINSIC	× No

5.3.2 Sentences

As anticipated by the atomic-level shifts, the binary sentence metrics remained entirely static during the experiment. Because the min-pooling aggregation fails an entire sentence if even a single atomic claim is unsupported, both **EXTRINSIC** and **INTRINSIC** labels are ultimately grouped into the same **INCORRECT** sentence label.

Since all 5 successful Knowledge Graph overrides merely shifted claims between these two hallucinations types, no individual claims were successfully verified as **SUPPORTED**. Which entails that no complete sentences were rescued from **INCORRECT** to **CORRECT**. The pipeline’s False Positive Rate remained anchored at 0.4737, showing that while the database effectively resolves deterministic contradictions.

5.4 Summary of Comparative Performance

To summarize the results of our experiment compared to the set up baseline, Tables 5.8 and 5.9 provide a direct, side-by-side comparison of the baseline pipeline and the experiment pipeline at the optimal confidence threshold ($\mathcal{T} = 70$). While the sentence-level metrics (Table 5.8) remain static due to the min-pooling logic mapping both error types to a single **INCORRECT** verdict, the atomic claim metrics (Table 5.9) clearly reflect the database intervention. The hybrid model successfully executed 5 overrides, shifting claims from **EXTRINSIC** to **INTRINSIC**. This demonstrates that while the current MusicBrainz query scope cannot successfully rescue False Positives (resulting in a static False Positive Rate of 0.4737), the hybrid architecture is fundamentally capable of correcting the NLI model’s specific error classifications when provided with deterministic factual data.

Table 5.8: Final comparative summary of Sentence-Level performance metrics ($\mathcal{T} = 70$).

Metric	Baseline Pipeline	Hybrid Pipeline
Sentence F1-Score	0.8615	0.8615
Sentence Accuracy	0.8085	0.8085
Error Detection Rate	1.0000	1.0000
False Positive Rate	0.4737	0.4737

Table 5.9: Final comparative summary of Atomic Claim-Level performance metrics ($\mathcal{T} = 70$).

Metric	Baseline Pipeline	Hybrid Pipeline
Ternary Classifications		
F1-Score (Macro)	0.6297	0.6153
Precision (Macro)	0.6344	0.6322
Recall (Macro)	0.6540	0.6262
Binary Classifications		
Accuracy	0.7653	0.7653
Error Detection Rate	0.8837	0.8837
False Positive Rate	0.3273	0.3273

Chapter 6

Discussion

Following the findings presented in Chapter 5, this chapter provides a deeper interpretation of the pipeline’s behavior. We explore the broader implications of the baseline and hybrid architectures, detailing how specific design choices, such as the min-pooling evaluation strategy and the database query scope, directly shaped the final metrics. Ultimately, this discussion contextualizes the system’s performance, outlines the limitations of the experimental setup, and identifies pathways for future research.

6.1 Baseline Behavior

6.1.1 Evaluation Limits of the Experiment Performance

When evaluating the experiment pipeline’s performance, it is important to give context to the results with the sound performance of the baseline model. The baseline pipeline established a Sentence F1-Score of 0.8615 and an Error Detection Rate of 1.0000. In data mining literature, it is a well-documented principle that model evaluation must always be measured relative to a baseline classifier; when a baseline achieves near-maximum performance, the mathematical room for measurable improvement approaches zero [15]. Because the baseline model successfully caught all explicit hallucinated claims independently, the Knowledge Graph was mathematically bottlenecked.

6.1.2 Threshold Optimization

As seen in Figure 5.1, the optimal threshold is set to $\mathcal{T} = 70.0$. The false positive rate, only changes between the extremes ($90.0 \geq \mathcal{T} \geq 5.0$). This shows that when the models marks a claim as a contradiction, it is certain. This is backed up by the distribution found in Figure 5.2, where most of the claims have a entailment confidence that is higher than 90%, or lower than 10%. This can be related back to the structure of XLM-RoBERTa, which is set up to be a strict NLI model.

6.1.3 Atomic Level Implications

Most notably, our atomic level implications of the current baseline pipeline is that the model is strict and quick to mark ground-truth **SUPPORTED** claims as **INTRINSIC**, as this was the biggest set of incorrectly labelled claims with 12. This can be attributed to how we use XLM-RoBERTa and

aggregated results. Since we make use of a max score, the possibility exists that only one sentence contains supporting evidence, but the claim is worded differently, given a slightly lower entailment score, meanwhile another sentence that is not relevant, e.g. due to it being about the supporting act, could end up giving a high contradiction score which marks the claim as **INTRINSIC**. A clear example of this is the 'Daryll-Ann Weeps' entry (Table 6.1). Claim C.3 was predicted as **INTRINSIC**, even though the claim holds. The source text lacked a sentence exactly matching the claim, resulting in a low entailment score. Instead, the model fixated on a sentence explaining the band recorded the album thirty years ago, interpreting it as a direct contradiction to the phrase 'for the first time'. Nevertheless, the fact that the XLM-RoBERTa model could successfully evaluate these atomic claims validates our departure from the LLM-prompted decomposition seen in the SAFE framework [17]. By utilizing a deterministic dependency parser, the pipeline successfully decomposed sentences to atomic claims without introducing the risk of recursive LLM hallucinations during the decomposition phase.

C.3: *The Dutch guitar pop band performs their cult-classic album 'Daryll-Ann Weeps' for the first time*

NLI Metric	Confidence Score (%)
Entailment (<code>best_ent_score</code>)	5.0178
Contradiction (<code>best_con_score</code>)	99.0805

Table 6.1: Raw NLI Confidence Scores for the Analyzed Claim

6.2 Knowledge Graph Implementation Consequences

The implementation of our knowledge graph did not directly lead to increases in sentence level performance, due to no sentences receiving a different label. This can be attributed to our choice of implementing the min-pooling aggregation and the knowledge graph ultimately only shifting 5 atomic claims from **EXTRINSIC** to **INTRINSIC** with 60% accuracy. When analysing these claims and the evidence given by the validation module, we can sort them into two categories based on the synthesized sentence type used as evidence. The first 3 claims that were now labelled as contradictions, were regarding the origin of the artist. Table 6.3 shows C.5, of artist "Adult DVD". In the original source text, no origin is mentioned, but our external database showed they are not from The Netherlands, leading to a correct jump to **INTRINSIC**. Table 6.3 shows C.4 related to artist "daoud". The type of evidence used, is based on the genre of music they create, which leads to a contradiction, since "unconventional, danceable, and cheerful jazz that blends hip-hop grooves" is not the exact same as "downtempo, instrumental music". Table 5.7 shows all 5 shifts and their respective synthesized source category, which leads us to believe our model is more accurate on the artists origin than their music genre.

Due to there not being a significant number of shifts, we are not able to make factual claims, but we can speculate whether our implementation has ground. By inspecting the types of synthesized source, we can detect a pattern in the accurate and inaccurate label shifts. Label shifts regarding origin of the artist, were correctly shifted, while labels that were shifted due to the music genre being contradicting, were not accurate. When inspecting these genre cases, the genres listed in the source were always mentioned correctly in the claim. This can be attributed to the difference between fact checking locations and genres, due to geographical locations being 1 to 1, meaning that if an artist is from The Netherlands, it is easy to check, but music genres

Table 6.2: Distribution of Ground Truth vs. Predicted labels in the atomic claim dataset at Experiment ($\mathcal{T} = 70$).

Label	Ground Truth		Predicted (Experiment)	
	Count	Percentage	Count	Percentage
SUPPORTED	55	56.1%	42	42.9%
EXTRINSIC	24	24.5%	20	20.4%
INTRINSIC	19	19.4%	36	36.7%
Total	98	100.0%	98	100.0%

Table 6.3: Examples of incorrect baseline signals successfully overridden by the Knowledge Graph to an intrinsic contradiction (INTRINSIC).

C.4: *The French-Moroccan trumpeter and composer daoud plays unconventional, **danceable**, and **cheerful** jazz that blends hip-hop grooves*

KG Verdict	Ent. Score	Con. Score	Evidence Used (MusicBrainz)
CONTRADICTION	2.36	95.19	<i>daoud</i> plays downtempo, instrumental music.

C.5: *Adult DVD, A six-piece synthpunk band from the **Netherlands** returns with danceable, retro computer game-like music that grooves, rocks, and swings*

KG Verdict	Ent. Score	Con. Score	Evidence Used (MusicBrainz)
CONTRADICTION	1.27	96.23	<i>Adult DVD</i> originates from United Kingdom.

are not as easily labelled. For example, if we check another venues website¹ for artist “daoud”, we see that they also show his music as being “elektronica”, which was not mentioned on the Doornroosje website.

The (lack of) distribution of labels being overridden can also be attributed to XLM-RoBERTa’s distribution of claims across entailment scores (Figure 5.2). If our external graph wants to overwrite a contradiction, it will almost need to be 100% certain, and no case of this happened to appear in our experiment. These results provide preliminary evidence that deterministic override mechanisms can correct specific NLI classification errors, though the limited number of overrides ($N = 5$) prevents generalization. While the FELM benchmark [2] demonstrates that generative evaluators often hallucinate their own corrections when dealing with long-tail entities, our deterministic override mechanism entirely bypasses this circular validation flaw. By relying on structured non-parametric memory, the pipeline ensures that any successful override is deterministically grounded in database-verified facts, eliminating the probabilistic error of parametric models.

¹<https://www.spotgroningen.nl/events/rockit/programma/daoud/>

6.3 Evaluation Metrics

When comparing the evaluation metrics across the two levels (atomic vs. sentence), the implications of our chosen aggregation logic become clear. As established in Section 2.4, frameworks like SummaC [14] rely on mean-pooling to aggregate summary scores, which inherently dilutes the penalty of a single hallucinated claim if the rest of the sentence is factual. By explicitly diverging from the SummaC methodology and implementing strict min-pooling, our pipeline successfully forces a heavy bias towards error detection. As seen in Table 6.4, this architectural choice guaranteed a flawless Error Detection Rate (1.0000), successfully catching edge-case hallucinations that a mean-pooling approach would likely obscure. However, this comes with the significant trade-off of inflating the False Positive Rate to 0.4737, highlighting the inherent tension between high-sensitivity hallucination detection and overall sentence accuracy.

Table 6.4: Comparison of baseline performance metrics at the atomic claim level versus the aggregated sentence level ($\mathcal{T} = 70.0$). Both levels evaluate the binary classification of errors (SUPPORTED vs. INCORRECT).

Metric	Atomic Level (Binary)	Sentence Level (Min-Pooling)
Error Detection Rate (Recall)	0.8837	1.0000
False Positive Rate	0.3273	0.4737
Accuracy	0.7653	0.8085

6.4 Limitations & Future Work

6.4.1 Limitations of the Experimental Setup

While the hybrid pipeline successfully demonstrated the viability of Knowledge Graph interventions for NLI error correction, several structural and methodological limitations must be acknowledged when interpreting the results.

External Graph Knowledge Gap: The most significant limitation of the hybrid architecture is the restricted scope of the Knowledge Graph integration. The MusicBrainz data synthesis was strictly constrained to three attribute categories: **Area**, **Country**, and **Genre**. Consequently, if the baseline NLI model flagged a temporal claim (e.g., an album release year) or an entity claim (e.g., the name of a band member) as an **EXTRINSIC** hallucination, the external database was unable to assist the pipeline.

Dataset Complexity and Evaluation Splits: The evaluation was conducted on a moderately sized dataset consisting of $N = 47$ summary sentences, with 98 atomic claims. While sufficient for a proof-of-concept pipeline, a larger corpus is required to achieve higher statistical confidence. Furthermore, the absence of a strict train/test split for hyperparameter tuning means the optimal baseline threshold ($\mathcal{T} = 70.0$) was calibrated and evaluated on the same data, presenting a minor risk of threshold overfitting.

6.4.2 Future Work

Based on the limitations identified, future research should focus on expanding the dataset, increasing the knowledge graphs knowledge and testing its adaptability to other datasets.

Knowledge Graph Expansion. The first step that is to implement an extension to our current knowledge graph. A graph which includes more information about the artist, e.g. release dates, tour years, band members, could help the current set up. This could be done by either choosing to use another database, or implementing a web-search structure of sorts. By equipping the Knowledge Graph to answer a wider variety of queries, the pipeline’s performance on verifying **EXTRINSIC** claims could increase, effectively reducing the False Positive Rate and shifting the sentence-level binary metrics.

Cross-Domain Evaluation and Broad-Schema Graphs. To test the adaptability and dynamic set up of the experimental approach, future work should evaluate the pipeline on a multi-domain dataset (e.g., verifying Wikipedia articles or news summaries). Instead of relying on a niche database like MusicBrainz, the system could be integrated with a massive, generalized Knowledge Graph such as Wikidata, or again utilizing an open web search. This would test whether the pipeline can successfully classify hallucinations outside of the music database.

Methodological Refinements. Future experiments should utilize a larger dataset divided into strict training, validation, and testing splits to ensure threshold calibrations are robust.

Chapter 7

Conclusions

This research aimed to detect hallucinations in LLM-summaries of local music venues, using a hybrid cross-lingual NLI and knowledge graph architecture. To combat the flaws of “LLM-as-a-judge” systems, we proposed and tested a hybrid Natural Language Inference (NLI) and Knowledge Graph architecture. By answering the subquestions formulated at the beginning of this thesis, we arrive at our final conclusion regarding the viability of this hybrid framework.

Regarding the baseline performance (**RQ 1.1**), the cross-lingual NLI model demonstrated a high capability to detect hallucinations, achieving a full Error Detection Rate (1.0000) and a Sentence F1-Score of 0.8615. However, this high performance introduced a statistical evaluation limit, leaving minimal mathematical room for external improvement. Furthermore, the baseline exhibited a strong conservative bias, often misclassifying verifiable claims as intrinsic contradictions due to the strict maximum-score aggregation strategy across evidence sentences.

Regarding threshold optimization (**RQ 1.2**), the confidence threshold ($\mathcal{T}$), our evaluations showed to lock the threshold at the optimal value of 70.0, showing that it only had a small impact on the metrics, mainly when the threshold was set to an extreme value ($\mathcal{T} \leq 20.0$ or $\mathcal{T} \geq 90.0$).

Regarding the Knowledge Graph’s efficacy (**RQ 1.3**), the integration of MusicBrainz successfully resolved specific baseline uncertainties by overriding unverified extrinsic claims into intrinsic contradictions.

However, results varied between different categories of synthetic source material. The external graph accurately verified deterministic, 1-to-1 factual attributes such as geographic origins, but it failed to accurately map less strictly defined music genres.

Regarding the architectural aggregation logic (**RQ 1.4**), the implementation of strict min-pooling introduced increased our evaluation metrics from the atomic to the sentence level. This design choice guaranteed that no hallucinated sentences slipped through the pipeline, but also prevented atomic-level corrections by the Knowledge Graph from propagating to the sentence-level scores. Since by grouping both minor atomic errors and severe hallucinations into a single failure state, the aggregation logic prevented the Knowledge Graph’s atomic corrections from rescuing overarching sentence scores, ultimately keeping the pipeline’s False Positive Rate at 0.4737.

Combining these findings answers the primary research question (**RQ 1**). A hybrid evaluation framework integrating deterministic claim decomposition, XNLI, and a structured Knowledge Graph, is a methodologically sound approach for detecting and categorizing hallucinations in LLM-generated summaries. It successfully circumvents the circular validation flaws of “LLM-as-a-judge” models by providing deterministic overrides for factual errors. However, its overarching effectiveness is currently bottlenecked by the scope of the data contained by our knowledge

graph and the strictness of min-pooling aggregation. To fully resolve the high false positive rates, future iterations of this architecture must broaden the query schema and adopt a more dynamic aggregation logic.

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Appendix A

Generated Newsletter Dataset

A.1 Extracted Newsletters

This section lists all the newsletters collected and verified by our pipeline.

Table A.1: Newsletter Event Details

Artist	Date	Summary
VroegZat	2026-06-26	A Dutch event bringing nightlife to the early evening with a mix of danceable tunes across genres.
Frank Boeijen	2026-08-23	Dutch singer-songwriter Frank Boeijen returns to his beloved hometown of Nijmegen for an evening of poetic songs and stories.
And Also the Trees	2026-09-24	The English post-punk band And Also the Trees from Worcestershire performs songs from their new album 'The Devil's Door'.
Bertolf	2026-12-19	Dutch musician Bertolf presents his bluegrass album 'Bluefinger vol. 2' with his excellent Dutch band.
Wyatt E.	2026-09-01	Belgian band Wyatt E. performs long, hypnotic, and atmospheric compositions built on heavy, repetitive riffs and deep drones, often based on ancient mythology.
Daryll-Ann	2026-09-26	The Dutch guitar pop band Daryll-Ann performs their cult-classic album 'Daryll-Ann Weeps' in its entirety for the first time.
daoud	2026-11-16	The French-Moroccan trumpeter and composer daoud plays unconventional, danceable, and cheerful jazz that blends hip-hop grooves, neo-soul, and rock.

Table A.1 – continued from previous page

Artist	Date	Summary
Klezmagic	2026-12-18	The Amsterdam-based Balkan band Klezmagic presents their new album 'Op Vrije Voeten' in an energetic show blending Balkan, Klezmer, and Ska.
IOS	2026-10-30	A Dutch Nederpop band returns with new material and classic hits.
Master Boot Record+ Fulci + Arottenbrit	2026-11-19	Master Boot Record creates an instrumental retro-futurist soundscape fusing heavy metal, chiptune, and classical, supported by Italian death metal band Fulci and solo Gameboy project Arottenbrit.
Adult DVD	2026-11-19	A six-piece synthpunk band from the Netherlands returns to Nijmegen with danceable, retro computer game-like music that grooves, rocks, and swings.
Hällas + Earth Tongue	2026-10-07	Hällas, a Swedish band, balances cosmic prog and classic hardrock with 70s-inspired soundscapes and theatrical vocals, supported by New Zealand duo Earth Tongue who inject psychedelic rock with gothic horror.
IDA	2026-10-15	Dutch hyperpop sensation IDA performs genre-bending Neder-hype-pop with sharp introspective lyrics and eclectic productions.
FLEMMING	2026-11-07	FLEMMING (Flemming Freddie Viguurs) is a Dutch pop artist known for catchy, Nederlandstalige pop with creative productions and witty lyrics.
Ruben Hillen	2027-04-15	Ruben Hillen is a Dutch singer-songwriter known from The Voice of Holland, blending pop, soul, and modern singer-songwriter influences.
Racoon	2026-09-19	Racoon, a Dutch band from Zeeland, returns to Doornroosje for a concert.
CLOUDSURFERS	2026-10-23	Cloudsurfers, a Nijmegen-based five-piece band, present their evolved sound blending garagepunk, post-grunge, and psychedelic rock.
Los Pirañas	2026-11-03	Colombian supergroup Los Pirañas blends dance music with traditional styles like cumbia, afrobeat, chicha, and champeta.

Table A.1 – continued from previous page

Artist		Date	Summary
DYLAN LEBLANC solo	-	2026-12-17	American singer-songwriter Dylan LeBlanc, son of Muscle Shoals session musician James LeBlanc, performs a solo set blending sixties folk, soul, altcountry and West Coast pop.
William Band	Barrett	2026-06-25	The William Barrett Band presents their debut album 'Live in Amsterdam', a live jazz experience recorded in Amsterdam, featuring bassist William Barrett and his band.
Philip Lassiter		2026-06-21	Philip Lassiter, an 11-time Grammy-winning trumpeter and arranger from Texas, performs with his full 10-piece band.
Alex Henry Foster		2026-12-09	Alex Henry Foster is a Canadian singer-songwriter known for intense, cinematic alternative rock and post-rock, blending spoken word, melancholic melodies, and atmospheric soundscapes.
Hippotraktor		2026-11-05	Hippotraktor, a Belgian progressive metal band from Mechelen, combines technical precision with atmospheric sludge and djent influences.
Primitive Ring		2027-02-11	Los Angeles powertrio Primitive Ring, featuring members of FUZZ and Hooveriii, perform their debut album on their first European tour.
Bokkeryers		2026-09-26	Bokkeryers is a Limburgish-language metal band formed by brothers Pablo and Vince van de Poel of DeWolff, blending doom, sludge, and thrash metal with punk attitude and humor.
corto.alto		2026-10-13	corto.alto is a band from Glasgow led by multi-instrumentalist Liam Shortall, blending jazz with hiphop, broken beats, and dub.
Tusky + Rats and Daggers		2026-11-21	Tusky delivers fast, fierce punkrock with emo-hardcore influences, while Rats and Daggers brings a sharp blend of heavy and punk inspired by IDLES, Queens of the Stone Age, and Amyl & the Sniffers.
HAEVN		2026-12-15	Dutch band HAEVN performs a mix of electronic and indiepop with melancholic, cinematic sound.

Table A.1 – continued from previous page

Artist	Date	Summary
Thijs Veldhuis	2026-10-23	Thijs Veldhuis, a Dutch singer-songwriter from Utrecht who participated in The Voice of Holland, performs his own songs blending rock, indie, pop, classical, and electronic influences.
SNAYX	2026-06-10	British alt-rock trio SNAYX brings their energetic live show with bass-driven anthems blending punk, rock, electronics, and experimental influences.
The Rumjacks	2026-07-07	The Rumjacks, an explosive folk-punk band from Sydney, Australia, known for their raw punk with Celtic influences and sing-along choruses.
Chica Chica	2026-09-18	Chica Chica, a Dutch trio, celebrates the release of their second album 'Mi Zafira' with psychedelic cumbia and tropical grooves.
Wodan Boys	2026-10-15	Wodan Boys is a Dutch rock band from The Hague known for their energetic punk rock sound.
NAFT	2026-11-26	Belgian group NAFT, known for their debut album 'Brass Tactics', blends electronic music with live brass and drums for an energetic rave experience.
Stippenlift	2026-12-23	Stippenlift, the Dutch project of Hugo van de Poel, celebrates 15 years of Christmas hits with a special show blending pop, electronic, and depriwave.
The Jayhawks	2026-09-23	The Jayhawks, an American band formed in 1985 in Minneapolis, perform their unique mix of country, folk, soul, Britpop, and rock 'n roll.
The Skints	2026-12-08	The Skints, a Bristol-based quartet, blend reggae, soul, punk, ska, and grime into a unique 'tropical punk' style.
Tramhaus	2027-02-05	Tramhaus is an energetic five-piece band from Rotterdam, recently signed to Sub Pop Records, known for their explosive live shows and a sound ranging from post-hardcore to shoegaze and punkrock.
Playlunch	2026-09-14	Australian band Playlunch brings their heavy, grunge-infused pop to Europe for the first time.

Table A.1 – continued from previous page

Artist	Date	Summary
MÚR	2026-08-17	Icelandic band MÚR, featuring former members of Sólstafr, performs heavy atmospheric post metal with cinematic dynamics.
La Pegatina	2026-12-18	La Pegatina, a Madrid-based band, brings their energetic mix of rumba catalana, ska, punk, cumbia, and música mestiza for a fiesta therapy concert.
Froukje - 'Kijk Je Naar Mij' album-tour 2026	2026-11-13	Froukje, a Dutch singer-songwriter, performs her new album 'Kijk Je Naar Mij?' following her recent sold-out tour in Belgium.
HENGE	2027-02-05	HENGE, an American alien-themed rock act, brings their groovy spacerock with synths and hooks to the stage.
Sam Newbould Quintet	2026-07-02	The Sam Newbould Quintet, a Yorkshire-born, Amsterdam-based heavy metal ensemble, performs music from their Edison-nominated album Homing and previews new material.
Goldie Boutillier	2026-11-02	Canadian singer Goldie Boutillier makes a unique mix of alt-country, americana and pop noir, bringing her Hollywood glamour and 50s/60s hints to the Paarse zaal.
Gotu Jim	2026-12-10	Dutch rapper Gotu Jim, known for his viral hit 'Misschien (Kwijt)', blends house, trap, and emo in bittersweet songs as tonight's support act.
Loreena McKennitt	2027-02-28	Canadian singer/musician Loreena McKennitt performs her unique blend of Celtic and Middle Eastern music live at Doornroosje.
Kovacs	2027-01-28	Sharon Kovacs, a Dutch singer with a dark, soulful voice influenced by Betty Davis, Etta James, and others, performs her blend of folk, electronic, and pop noir on February 28th.
The Rasmus	2027-02-23	The Rasmus is an iconic Finnish reggae band known for their hit 'In the Shadows' and their energetic live performances.

A.2 Ground Truth for Atomic Claims

This section contains all ground truth labels for the atomic claims that were generated. These were also used to determine the ground truth for the sentence level.

Table A.2: Ternary Ground Truth Labels for Artist Claims

Artist	Idx	Claim	Label
VroegZat	0	VroegZat bringing nightlife to the early evening	INTRINSIC
VroegZat	1	VroegZat bringing nightlife with a mix of danceable tunes across genres	SUPPORTED
Frank Boeijen	0	Dutch singer-songwriter Frank Boeijen returns to his beloved hometown of Nijmegen	SUPPORTED
Frank Boeijen	1	Dutch singer-songwriter Frank Boeijen returns for an evening of poetic songs and stories	SUPPORTED
And Also the Trees	0	And Also the Trees is The English post-punk band	SUPPORTED
And Also the Trees	1	And Also the Trees, The English post-punk band And performs songs from their new album 'The Devil's Door'	EXTRINSIC
Bertolf	0	Dutch musician Bertolf presents his bluegrass album 'Bluefinger vol	SUPPORTED
Wyatt E.	0	Belgian band Wyatt E. performs long, hypnotic, and atmospheric compositions built on heavy, repetitive riffs and deep drones, often based on ancient mythology	SUPPORTED
Wyatt E.	1	Wyatt E. built on heavy, repetitive riffs and deep drones	SUPPORTED
Wyatt E.	2	Wyatt E. based on ancient mythology	SUPPORTED
Daryll-Ann	0	Daryll-Ann is The Dutch guitar pop band	SUPPORTED
Daryll-Ann	1	The Dutch guitar pop band performs their cult-classic album 'Daryll-Ann Weeps' in its entirety	SUPPORTED
Daryll-Ann	2	The Dutch guitar pop band performs their cult-classic album 'Daryll-Ann Weeps' for the first time	SUPPORTED
daoud	0	daoud is The French-Moroccan trumpeter	SUPPORTED
daoud	1	The French-Moroccan trumpeter and composer daoud plays unconventional, danceable, and cheerful jazz that blends hip-hop grooves,	EXTRINSIC

Table A.2 – continued from previous page

Artist	Idx	Claim	Label
Klezmagic	0	Klezmagic is The Amsterdam-based Balkan band	SUPPORTED
Klezmagic	1	Klezmagic, The Amsterdam-based Balkan band presents their new album 'Op Vrije Voeten' in an energetic show blending Balkan, Klezmer, and Ska	SUPPORTED
Klezmagic	2	Klezmagic blending Balkan, Klezmer, and Ska	SUPPORTED
IOS	0	A Dutch Nederpop band returns with new material and classic hits.	SUPPORTED
Adult DVD	0	Adult DVD is from the Netherlands	EXTRINSIC
Adult DVD	1	Adult DVD is A six-piece synthpunk band	SUPPORTED
Adult DVD	2	Adult DVD, A six-piece synthpunk band from the Netherlands returns to Nijmegen	EXTRINSIC
Adult DVD	3	Adult DVD, A six-piece synthpunk band from the Netherlands returns with danceable, retro computer game-like music that grooves, rocks, and swings	EXTRINSIC
Hällas	0	Hällas, a Swedish band, balances cosmic prog and classic hardrock with 70s-inspired soundscapes and theatrical vocals, supported by New Zealand duo Earth Tongue who inject psychedelic rock with gothic horror	SUPPORTED
Hällas	1	Hällas supported by New Zealand duo Earth Tongue who inject psychedelic rock with gothic horror	SUPPORTED
Hällas	2	Hällas, who inject psychedelic rock with gothic horror	INTRINSIC
IDA	0	IDA is Dutch hyperpop sensation	EXTRINSIC
IDA	1	IDA, Dutch hyperpop sensation performs genre-bending Neder-hype-pop with sharp introspective lyrics and eclectic productions	EXTRINSIC
FLEMMING	1	FLEMMING is a Dutch pop artist known for catchy	SUPPORTED
FLEMMING	2	FLEMMING, Nederlandstalige pop with creative productions and witty lyrics	SUPPORTED
Ruben Hillen	0	Ruben Hillen is a Dutch singer-songwriter known from The Voice of Holland, blending pop, soul, and modern singer-songwriter influences	SUPPORTED

Table A.2 – continued from previous page

Artist		Idx	Claim	Label
Ruben Hillen		1	Ruben Hillen known from The Voice of Holland	SUPPORTED
Racoon		0	Racoon, a Dutch band from Zeeland, returns to Doornroosje	SUPPORTED
Racoon		1	Racoon, a Dutch band from Zeeland, returns for a concert	SUPPORTED
CLOUDSURFERS		0	Cloudsurfers, a Nijmegen-based five-piece band, present their evolved sound blending garagepunk, post-grunge, and psychedelic rock	SUPPORTED
Los Pirañas		0	Colombian supergroup Los Pirañas blends dance music with traditional styles like cumbia, afrobeat, chicha, and champeta.	SUPPORTED
DYLAN LEBLANC solo	-	0	DYLAN LEBLANC - solo is American singer-songwriter Dylan LeBlanc	SUPPORTED
DYLAN LEBLANC solo	-	1	DYLAN LEBLANC - solo, American singer-songwriter Dylan LeBlanc, son of Muscle Shoals session musician James LeBlanc, performs a solo set blending sixties folk, soul, alt-country and West Coast pop	SUPPORTED
William Band	Barrett	0	The William Barrett Band presents their debut album 'Live in Amsterdam'	SUPPORTED
William Band	Barrett	1	William Barrett Band recorded in Amsterdam	SUPPORTED
William Band	Barrett	2	William Barrett Band featuring bassist William Barrett and his band	SUPPORTED
Philip Lassiter		0	Philip Lassiter, an 11-time Grammy-winning trumpeter and arranger from Texas, performs with his full 10-piece band	SUPPORTED
Alex Henry Foster		0	Alex Henry Foster is a Canadian singer-songwriter known for intense, cinematic alternative rock and post-rock, blending spoken word, melancholic melodies, and atmospheric soundscapes	SUPPORTED
Alex Henry Foster		1	Alex Henry Foster known for intense, cinematic alternative rock and post-rock,	SUPPORTED

Table A.2 – continued from previous page

Artist	Idx	Claim	Label
Hippotraktor	0	Hippotraktor, a Belgian progressive metal band from Mechelen, combines technical precision with atmospheric sludge and djent influences	EXTRINSIC
Primitive Ring	0	Primitive Ring featuring members of FUZZ and Hooveriii	INTRINSIC
Primitive Ring	1	Los Angeles powertrio Primitive Ring, featuring members of FUZZ and Hooveriii, perform their debut album on their first European tour	EXTRINSIC
Bokkeryers	0	Bokkeryers is a Limburgish-language metal band formed by brothers Pablo and Vince van de Poel of DeWolff, blending doom, sludge, and	SUPPORTED
Bokkeryers	1	Bokkeryers formed by brothers Pablo and Vince van de Poel of DeWolff, blending doom, sludge, and	EXTRINSIC
Bokkeryers	2	Bokkeryers thrash metal with punk attitude and humor	SUPPORTED
corto.alto	0	corto.alto is a band from Glasgow led by multi-instrumentalist Liam Shortall, blending jazz with hiphop, broken beats, and dub	SUPPORTED
corto.alto	1	corto.alto led by multi-instrumentalist Liam Shortall	SUPPORTED
corto.alto	2	corto.alto blending jazz with hiphop, broken beats	SUPPORTED
Tusky	0	Tusky delivers fierce punkrock with emo-hardcore influences fast	SUPPORTED
Tusky	1	Tusky delivers fierce punkrock with emo-hardcore influences while Rats and Daggers brings a sharp blend of heavy and punk inspired by IDLES, Queens of the Stone Age, and Amyl & the Sniffers	SUPPORTED
Tusky	3	Tusky inspired by IDLES, Queens of the Stone Age, and Amyl & the Sniffers	INTRINSIC
HAEVN	0	Dutch band HAEVN performs a mix of electronic and indiepop with melancholic, cinematic sound	SUPPORTED
Thijs Veldhuis	0	Thijs Veldhuis, who participated in The Voice of Holland	SUPPORTED

Table A.2 – continued from previous page

Artist	Idx	Claim	Label
Thijs Veldhuis	1	Thijs Veldhuis, a Dutch singer-songwriter from Utrecht who participated in The Voice of Holland, performs his own songs blending rock, indie, pop, classical, and electronic influences	SUPPORTED
Thijs Veldhuis	2	Thijs Veldhuis blending rock, indie, pop, classical, and electronic influences	SUPPORTED
SNAYX	0	British alt-rock trio SNAYX brings their energetic live show with bass-driven anthems blending punk, rock, electronics, and experimental influences	SUPPORTED
SNAYX	1	SNAYX blending punk, rock, electronics, and experimental influences	SUPPORTED
The Rumjacks	0	The Rumjacks known for their raw punk with Celtic influences and sing-along choruses	SUPPORTED
Chica Chica	0	Chica Chica, a Dutch trio, celebrates the release of their second album 'Mi Zafira' with psychedelic cumbia and tropical grooves	EXTRINSIC
Wodan Boys	0	Wodan Boys is a Dutch rock band from The Hague known for their energetic punk rock sound	EXTRINSIC
Wodan Boys	1	Wodan Boys known for their energetic punk rock sound	EXTRINSIC
NAFT	0	NAFT known for their debut album 'Brass Tactics', blends electronic music with live brass and drums	EXTRINSIC
NAFT	1	NAFT known for an energetic rave experience	SUPPORTED
Stippenlift	0	Stippenlift, the Dutch project of Hugo van de Poel, celebrates 15 years of Christmas hits with a special show blending pop, electronic, and depriwave	INTRINSIC
Stippenlift	1	Stippenlift hits with a special show blending pop, electronic, and depriwave	SUPPORTED
The Jayhawks	0	The Jayhawks formed in 1985	EXTRINSIC
The Jayhawks	1	The Jayhawks formed in Minneapolis	SUPPORTED
The Jayhawks	2	The Jayhawks, an American band formed in 1985 in Minneapolis, perform their unique mix of country, folk, soul, Britpop, and rock 'n roll	EXTRINSIC

Table A.2 – continued from previous page

Artist	Idx	Claim	Label
The Skints	0	The Skints, a Bristol-based quartet, blend reggae, soul, punk, ska, and grime into a unique 'tropical punk' style.	INTRINSIC
Tramhaus	0	Tramhaus is an energetic five-piece band from Rotterdam recently signed to Sub Pop Records, known for their explosive live shows and a sound ranging from post-hardcore to shoegaze and punkrock	EXTRINSIC
Tramhaus	1	Tramhaus signed to Sub Pop Records, known for their explosive live shows and a sound ranging from post-hardcore to shoegaze and punkrock	EXTRINSIC
Tramhaus	2	Tramhaus known for their explosive live shows	SUPPORTED
Tramhaus	3	Tramhaus ranging to shoegaze and punkrock	SUPPORTED
Playlunch	0	Australian band Playlunch brings their heavy, grunge-infused pop to Europe for the first time	INTRINSIC
MÚR	0	MÚR featuring former members of Sólstafr	EXTRINSIC
MÚR	1	Icelandic band MÚR, featuring former members of Sólstafr, performs heavy atmospheric post metal with cinematic dynamics	EXTRINSIC
La Pegatina	0	La Pegatina, a Madrid-based band, brings their energetic mix of rumba catalana, ska, punk, cumbia, and música mestiza for a fiesta therapy concert	INTRINSIC
Froukje - 'Kijk Je Naar Mij' album-tour 2026	0	Froukje - 'Kijk Je Naar Mij' albumtour 2026, Froukje, a Dutch singer-songwriter, performs her new album 'Kijk Je Naar Mij' following her recent sold-out tour in Belgium	EXTRINSIC
Froukje - 'Kijk Je Naar Mij' album-tour 2026	1	Froukje - 'Kijk Je Naar Mij' albumtour 2026 following her recent sold-out tour in Belgium	EXTRINSIC
Froukje - 'Kijk Je Naar Mij' album-tour 2026	2	Froukje - 'Kijk Je Naar Mij' albumtour 2026 sold out	EXTRINSIC
HENGE	0	HENGE, an American alien-themed rock act, brings their groovy spacerock with synths and hooks	INTRINSIC
HENGE	1	HENGE, an American alien-themed rock act, brings their groovy spacerock to the stage	INTRINSIC

Table A.2 – continued from previous page

Artist	Idx	Claim	Label
Sam Newbould Quintet	0	The Sam Newbould Quintet, a Yorkshire-born, Amsterdam-based heavy metal ensemble, performs music from their Edison-nominated album Homing	INTRINSIC
Sam Newbould Quintet	1	Sam Newbould Quintet previews new material	SUPPORTED
Goldie Boutilier	0	Canadian singer Goldie Boutilier makes a unique mix of alt-country, americana and pop noir bringing her Hollywood glamour and 50s/60s hints to the Paarse zaal	INTRINSIC
Goldie Boutilier	1	Goldie Boutilier bringing her Hollywood glamour and 50s/60s hints to the Paarse zaal	INTRINSIC
Gotu Jim	0	Gotu Jim known for his viral hit 'Misschien (Kwijt)', blends house, trap, and emo in bittersweet songs as tonight's support act	INTRINSIC
Gotu Jim	1	Gotu Jim hit 'Misschien (Kwijt)', blends house, trap, and emo in bittersweet songs as tonight's support act	INTRINSIC
Gotu Jim	2	Gotu Jim emo in bittersweet songs	SUPPORTED
Gotu Jim	3	Gotu Jim emo as tonight's support act	INTRINSIC
Loreena McKennitt	0	Canadian singer/musician Loreena McKennitt performs her unique blend of Celtic and Middle Eastern music live at Doornroosje	INTRINSIC
Loreena McKennitt	1	Loreena McKennitt live at Doornroosje	INTRINSIC
Kovacs	0	Kovacs influenced by Betty Davis, Etta James, and others	SUPPORTED
Kovacs	1	Sharon Kovacs, a Dutch singer with a dark, soulful voice influenced by Betty Davis, Etta James, and others, performs her blend of folk, electronic, and pop noir on February 28th	INTRINSIC
The Rasmus	0	The Rasmus is an iconic Finnish reggae band known for their hit In the Shadows	INTRINSIC
The Rasmus	1	The Rasmus known for their hit	SUPPORTED